

Basic Computer Programming

1. History of C

C is a general-purpose high level language that was originally developed by Dennis Ritchie for the UNIX operating system. It was first implemented on the Digital Equipment Corporation PDP-11 computer in 1972.

The UNIX operating system and virtually all UNIX applications are written in the C language. C has now become a widely used professional language for various reasons.

- Easy to learn
- Structured language
- It produces efficient programs.
- It can handle low-level activities.
- It can be compiled on a variety of computers.
- Utilities

2. Importance of C:

- C is a structured programming language
- C supports functions that enables easy maintainability of code, by breaking large file into smaller modules
- Comments in C provides easy readability
- C is a powerful language
- Rich in built in functions and operators that can be used to write any complex program
- Programs written in C are efficient and fast
- C is highly portable

3. C: Characteristics

- C takes a middle path between low-level assembly language...
 - a. Direct access to memory layout through pointer manipulation
 - b. Concise syntax, small set of keywords
- A high-level programming language like Java:
 - c. Block structure
 - d. Some encapsulation of code, via functions
 - e. Type checking (pretty weak)

4. C: Dangers

- C is not object oriented!
 - a. Can't "hide" data as "private" or "protected" fields
 - b. You can follow standards to write C code that looks object-oriented, but you have to be disciplined – will the other people working on your code also be disciplined?
- C has portability issues
 - c. Low-level "tricks" may make your C code run well on one platform – but the tricks might not work elsewhere
- The compiler and runtime system will rarely stop your C program from doing stupid/bad things
 - d. Compile-time type checking is weak
 - e. No run-time checks for array bounds errors, etc. like in Java

Structured Programming

- ▶ There are three main principles of structured programming:
 1. Program Design using top-down or bottom-up approach.
 2. Decomposition of program into components
 3. Structuring of control flow

Main Features of Structured Programming Language

1. Division of Complex problems into small procedures and functions.
2. No presence of GOTO Statement
3. The main statement include – If-then-else, Call and Case statements.
4. Large set of operators like arithmetic, relational, logical, bit manipulation, shift and part word operators.
5. Inclusion of facilities for implementing entry points and external references in program.

Unstructured Programming Language

- ▶ A programming language in which the entire logic of the program is written as a single continuous (nonstop or unbroken) block is called "unstructured Programming".
- ▶ Program written by following this language is very difficult to modify and to debug

Why C language is Structured Language?

- ▶ In order to accomplish any task, C-language divide the problem into smaller modules called functions or procedure each of which handles a particular job. That is why C-language is also called as the structured programming language. The program which solves the entire problem is a collection of such functions.
- ▶ In C Programming when writing a program we add preprocessor directive `#include<filename>` , than functions like `main()` , then opening of braces `{ }` , inside the

braces there are two parts declaration of variables and execution of statements which is a step by step process and well structured so anyone in the first look can understand what codes are doing what in which section.

- ▶ The structured programming languages allows the program to be split into multiple blocks of execution.
- ▶ In C, you can split the program into such named blocks called functions.

That is why it is called a structured programming language

Features of C Language

C is the widely used language. It provides many **features** that are given below.

1. Simple
2. Machine Independent or Portable
3. Mid-level programming language
4. structured programming language
5. Rich Library
6. Memory Management
7. Fast Speed
8. Pointers
9. Recursion
10. Extensible

Structured Vs. Unstructured

Comparison between Structured and Unstructured Programming Language:

	Structured Programming Language	Unstructured Programming Language
Also known as	Modular Programming	Non-structured Programming
Purpose	To enforce a logical structure on the program being written to make it more efficient and easier to	Just to code.

	understand and modify.	
Programming	Divides the program into smaller units or modules.	The entire program must be coded in one continuous block.
Freedom	Has some limitations	Offers freedom to programmers to program as they want
Allowed data types	Structured languages allow a variety of data types.	Non-structured languages allow only basic data types, such as numbers, strings and arrays (numbered sets of variables of the same type).
Modify and debug	Easy to modify and to debug	Very difficult to modify and to debug
Languages	C, C+, C++, C#, Java, PERL, Ruby, PHP, ALGOL, Pascal, PL/I and Ada	early versions of BASIC (such as MSX BASIC and GW-BASIC), JOSS, FOCAL, MUMPS, TELCOMP, COBOL, machine-level code, early assembler systems (without procedural metaoperators), assembler debuggers and some scripting languages such as MS-DOS batch file language.

5. Program in C

C program basically has the following form:

- Preprocessor Commands
- Functions
- Variables
- Statements & Expressions
- Comments

The following program is written in the C programming language. Open a text file *hello.c* using vi editor and put the following lines inside that file.

```
#include <stdio.h>
int main()
{
    printf("Hello, World! \n");
    /* My first program */
    return 0;
}
```

Preprocessor Commands: These commands tell the compiler to do preprocessing before doing actual compilation. Like *#include <stdio.h>* is a preprocessor command which tells a C compiler to include *stdio.h* file before going to actual compilation. You will learn more about C Preprocessors in C Preprocessors session.

Functions: are main building blocks of any C Program. Every C Program will have one or more functions and there is one mandatory function which is called *main ()* function. This function is prefixed with keyword *int* which means this function returns an integer value when it exits. This integer value is returned using *return* statement.

The C Programming language provides a set of built-in functions. In the above example *printf()* is a C built-in function which is used to print anything on the screen. Check Builtin function section for more detail.

You will learn how to write your own functions and use them in Using Function session.

Variables: are used to hold numbers, strings and complex data for manipulation. You will learn in detail about variables in C Variable Types.

Statements & Expressions: Expressions combine variables and constants to create new values. Statements are expressions, assignments, function calls, or control flow statements which make up C programs.

Comments: are used to give additional useful information inside a C Program. All the comments will be put inside */*...*/* as given in the example above. A comment can span through multiple lines.